

Frédéric Andrian

AI Engineer · Generative AI, LLMs and Full-Stack Development

Bordeaux, France

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Professional Summary

Early-career AI Engineer completing an MSc-equivalent engineering degree in Artificial Intelligence and Cognitive Science. Built an end-to-end AI application at INRIA using Python, FastAPI, MongoDB, React, Next.js, the OpenAI API, and RAG. Experienced in framework-free agent development, LLM evaluation, quantization, Docker, and CI/CD. Seeking to build reliable AI products that connect user needs with practical backend, orchestration, and data components. Available from November 2026.

Experience

INRIA, Mnemosyne Team

Bordeaux, France

AI Research Engineer Intern

Mar. 2026 – Sep. 2026

- Independently designed and developed, with guidance from my supervisors, an end-to-end multi-step AI assistant that helps researchers discover scientific papers.
- Integrated the OpenAI API and built the agent without an orchestration framework to understand and control its tools, state, and multi-step execution mechanisms.
- Built Python and FastAPI backend services, REST APIs documented with OpenAPI, and MongoDB persistence.
- Developed a React and Next.js web interface that enabled researchers to interact with the AI assistant and its tools.
- Implemented a RAG pipeline using document chunking, embeddings, semantic search, and the ChromaDB vector database.
- Benchmarked Qwen, Gemma, GLM, and multiple quantization configurations on an HPC cluster with Slurm across latency, TTFT, response quality, and context size.
- Dockerized selected tools and implemented a GitHub CI/CD pipeline to automate integration.
- Managed a Dockerized web application that enabled users to interact with LLMs installed locally on a Mac Studio, including model installation and configuration.
- Developed a Python tool to automatically collect scientific paper metadata from multiple public sources.
- Implemented webhooks to relay the results of scheduled tasks.
- Evaluated OpenClaw as an agent platform within an isolated Docker environment.
- Defined an experimental protocol, analyzed results, and authored a scientific paper documenting the model evaluation.

Karunya University

Coimbatore, India

Reinforcement Learning Research Intern

Jun. 2025 – Sep. 2025

- Designed, implemented, and compared deterministic path planning with a reinforcement learning approach for autonomous navigation of a hospital robot.
- Deployed and optimized the software and reinforcement learning model on an embedded Raspberry Pi.

Machine Learning Project

League of Legends Player Assistant

Academic Project

Machine Learning Developer

- Collected match data through an API, cleaned and preprocessed the dataset, and trained a Random Forest model to help players make decisions during League of Legends matches.
- Integrated the model outputs into a Discord bot to make predictions and assistance accessible to players.

Technical Skills

Languages and Backend Generative AI and Agents

Python, Java, FastAPI, Node.js, REST APIs, OpenAPI

LLMs, OpenAI API, framework-free multi-step agents, tool design and orchestration, LangChain project experience

RAG and Data LLM Evaluation

Chunking, embeddings, semantic search, ChromaDB, MongoDB, SQL

Qwen, Gemma, GLM, quantization, latency, TTFT, response quality, context size

Frontend

React, Next.js

Engineering and DevOps Machine Learning

Linux, Docker, Git/GitHub, CI/CD, Slurm, HPC environments

PyTorch, Deep Learning, Reinforcement Learning, Random Forest, data preprocessing, NLP

Languages

French (native), English (professional, IELTS 7.5), Spanish (basic)

Education

ENSC, Bordeaux INP

Bordeaux, France

Engineering Degree, MSc-equivalent, in Artificial Intelligence and Cognitive Science

2023 – 2026

- Coursework in software engineering, machine learning, large language models, natural language processing, deep learning, and human-computer interaction.